Register

IO

Name	Size	Input/Output	Function
clk	1	input	clock
rst	1	input	Reset register
d	1	input	Data into the register
q	1	output	Output data
set	1	input	Pre-sets the data if we want to store data before

Reset

Reset is synchronous and signaled by setting rst high for at least 1 clock cycle. When the reset is complete, the register will be set LOW.

set

Reset is synchronous and signaled by setting set high for at least 1 clock cycle. When the set is complete, the register will be set HIGH or LOW based on the set data.

ROM

IO

Name	Size	Input/Output	Function

clk	1	input	clock
addr	1	input	Selecting the register to take data bits from
q	1	output	Output data

Address

We will read memory through a mux reading the address from the outputs of the shift registers.

System Verilog

```
module rom (
input logic clk,
input logic[15:0] addr,
output logic[47:0] q);
// Declare the ROM as an array
logic [47:0] rom[65536];
initial // Read the ROM contents from a file
begin
$readmemb("rom_init.txt", rom);
end
// Access the ROM
assign q = rom[addr];
endmodule
```